Package ‘SII’

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Type Package

Title Calculate ANSI S3.5-1997 Speech Intelligibility Index ('SII')

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Description Calculates ANSI S3.5-1997 Speech Intelligibility Index ('SII'), a standard method for computing the intelligibility of speech from acoustical measurements of speech, noise, and hearing thresholds. This package includes data frames corresponding to Tables 1 - 4 in the ANSI standard as well as functions utilizing these tables and user-provided hearing threshold and noise level measurements to compute the 'SII' score. The methods implemented here extend the standard computations to allow calculation of 'SII' when the measured frequencies do not match those required by the standard by applying interpolation. Furthermore, the package now includes advanced clinical methods to predict aided 'SII' based on hearing aid prescriptive rationales (such as 'NAL-NL2' and 'DSL'), calculate real-ear insertion gains, estimate maximum power output ('SSPL90'), and prescribe dynamic range compression settings. Development of this package was originally funded by the Center for 'Bioscience' Education and Technology ('CBET') of the Rochester Institute of Technology ('RIT').

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`SII-package` *Calculate ANSI S3.5-1997 Speech Intelligibility Index*

Description

This package calculates ANSI S3.5-1997 Speech Intelligibility Index (SII), a standard method for computing the intelligibility of speech from acoustical measurements of speech, noise, and hearing thresholds. This package includes data frames corresponding to Tables 1 - 4 in the ANSI standard as well as a function utilizing these tables and user-provided hearing threshold and noise level measurements to compute the SII score. The methods implemented here extend the standard computations to allow calculation of SII when the measured frequencies do not match those required by the standard by applying interpolation to obtain values for the required frequencies.

Author(s)

Gregory R. Warnes `<greg@warnes.net>`

References

ANSI S3.5-1997, "American National Standard Methods for Calculation of the Speech Intelligibility Index" American National Stan- dards Institute, New York.

Other software programs for calculating SII are available from https://sii.to/html/programs.html.

Examples

```
## Example C.1 from ANSI S3.5-1997 Annex C
sii.C1 <- sii(
              speech   = c(50.0, 40.0, 40.0, 30.0, 20.0,  0.0),
              noise    = c(70.0, 65.0, 45.0, 25.0,  1.0,-15.0),
              threshold= c( 0.0,  0.0,  0.0,  0.0,  0.0,  0.0),
              method="octave"
       )
sii.C1                        # rounded to 2 digits by default
print(sii.C1$sii, digits=20)  # full precision
summary(sii.C1)               # full details
plot(sii.C1)                  # plot
## The value given in the Standard is $0.504$.
```

calculate_loudness *Calculate Loudness (Sones)*

Description

Calculates the estimated loudness in sones for a given Speech Intelligibility Index (SII) object.

Usage

```
calculate_loudness(x)
```

Arguments

x	An object of class SII, representing the output of the sii() function.

Value

Returns a numeric value representing the total loudness in sones.

Author(s)

Mark Shaver

Examples

```
sii.res <- sii(speech=c(50, 40, 40, 30, 20, 0), threshold=rep(0,6), method="octave")
calculate_loudness(sii.res)
```

critical *Constants Tables for ANSI S3.5-1997 Speech Intelligibility Index (SII)*

Description

Tables of constants for ANSI S3.5-1997 Speech Intelligibility Index (SII)

Usage

```
data(critical)
data(equal)
data(onethird)
data(octave)
data(overall.spl)
```

Format

Each data frames has 6-21 observations and a subset of the following variables:

`fi` Center frequency of SII band, Hz

`li` Lower limit of frequency band, Hz

`hi` Upper limit of frequency band, Hz

`Deltai` Band width adjustment, dB

`Ii` Band importance function

`normal`**,** `raised`**,** `loud` **and** `shout` Standard spectrum levels for vocal effort levels "normal", "raised", "loud", and "shout", respectively, dB

`Xi` Spectrum level of internal noise, dB

`Fi` Band importance function (weight)

Details

These data objects provide constant tables 1 – 4 from the ANSI S3.5-1997.

`critical` Table 1: Critical band SII procedure constants

`equal` Table 2:Equally contributing (17 band) critical band SII

`onethird` Table 3: One-third octave band SII procedure constants

`octave` Table 4: Octave band SII procedure constants

`overall.spl` Overall sound pressure level (SPL) for the for vocal effort levels "normal", "raised", "loud", and "shout", in dB

Source

ANSI S3.5-1997, "American National Standard Methods for Calculation of the Speech Intelligibility Index" American National Standards Institute, New York.

References

ANSI S3.5-1997, "American National Standard Methods for Calculation of the Speech Intelligibility Index" American National Standards Institute, New York.

Examples

```
data(critical)
critical # show entire table

data(equal)
names(equal)
equal$fi # extract just the frequency band centers

data(onethird)
barplot(onethird$Ii) # plot band importance function (weights)

data(octave)
```

```
round(octave, digits=2) # just 2 digits

data(overall.spl)
overall.spl
```

`launch_app` *Launch the SII Interactive Dashboard*

Description

Launches a local Shiny web application that provides an interactive graphical interface to the SII calculation engine and plotting utilities.

Usage

```
launch_app()
```

Details

The interactive dashboard allows you to dynamically adjust hearing thresholds across standard clinical frequencies (250 Hz - 8000 Hz) using sliders. It provides real-time visualization of the Clinical SPLogram and the Insertion Gain / Compression plot based on the selected prescriptive rationale (e.g., Unaided, NAL-R, Open-NL) and desensitization settings.

Note: This function requires the `shiny` and `bslib` packages to be installed.

Author(s)

Gregory R. Warnes <`greg@warnes.net`>, Mark Shaver <`mark.shaver@posteo.net`>

See Also

`sii`, `plot.SII`

Examples

```
## Not run:
  # Launch the interactive SII dashboard in your web browser
  launch_app()

## End(Not run)
```

sic.critical *Alternative ANSI S3.5-1997 SII Transfer Function Weights*

Description

Alternative ANSI S3.5-1997 Speech Intelligibility Index (SII) transfer function weights for for various types of speech material.

Usage

```
data(sic.critical)
data(sic.onethird)
data(sic.octave)
```

Format

Each data frame contains the following 8 variables, each corresponding the the transfer function weights for a specific type of speech material:

`fi` Center frequency, Hz

`SII` Standard SII transfer function (weights)

`NNS` NNS (various nonsense syllable tests where most of the English phonems occur equally often)

`CID22` CID-W22 (PB-words)

`NU6` NU6 monosyllables

`DRT` DRT (Diagnostic Rhyme Test)

`ShortPassage` short passages of easy reading material

`SPIN` SPIN monosyllables

`CST` Connected Speech Test

Details

`sic.critical` provides alternative weights for the critical band SII procedure.

`sic.threeoctave` provides alternative weights for the one-third octave frequency band SII procedure.

`octave` provides alternative weights for the octave frequency band SII procedure.

note

There is no table of alternative weights for the equally-weighted SII band procedure as the weights for this method are (by definition) constant across all bands.

Source

All values except the CST columns are from:

ANSI S3.5-1997, "American National Standard Methods for Calculation of the Speech Intelligibility Index" American National Standards Institute, New York.

Values in the CST columns are from the original Connected Speech Test (CST) dataset.

References

ANSI S3.5-1997, "American National Standard Methods for Calculation of the Speech Intelligibility Index" American National Standards Institute, New York.

Examples

```
## Load the alternative weights for the critical band method
data(sic.critical)

## display the weights
round(sic.critical,3)

## draw a comparison plot
ngroup <- ncol(sic.critical)
matplot(x=sic.critical[,1], y=sic.critical[,-1],
        type="o",
        xlab="Frequency, Hz",
        ylab="Weight",
        log="x",
        lty=1:ngroup,
        col=rainbow(ngroup)
)
legend(
       "topright",
       legend=names(sic.critical)[-1],
       pch=as.character(1:ngroup),
       lty=1:ngroup,
       col=rainbow(ngroup)
       )

data(threeoctave)
data(octave)
```

---

sii *Compute ANSI S3.5-1997 Speech Intelligibility Index (SII)*

Description

Compute the Speech Intelligibility Index (SII) described by ANSI specification S3.5-1997, including extensions for conductive hearing loss. Optionally apply interpolation obtain values for the required frequencies.

Usage

```
sii(speech = c("normal", "raised", "loud", "shout"),
    noise, threshold, loss, freq,
    method = c("critical", "equal-contributing",
               "one-third octave", "octave"),
    importance = c("SII", "NNS", "CID22", "NU6", "DRT",
                   "ShortPassage", "SPIN", "CST"),
    interpolate=FALSE,
    prescription=NULL,
    desensitization=FALSE,
    ldl=NULL,
    gender="male",
    experience="experienced",
    config="bilateral",
    age="adult",
    age_years=NULL,
    age_months=NULL,
    coupling="custom_occluded",
    module="standard",
    transducer="inserts",
    custom_gain=NULL)
## S3 method for class 'SII'
print(x, digits=3, ...)
## S3 method for class 'SII'
plot(x, clinical=FALSE, legend=TRUE, legend_only=FALSE, ...)
## S3 method for class 'SII'
summary(object, digits=2, ...)
```

Arguments

`speech`	Either a numeric vector providing E_i', the equivalent speech spectrum level (in dB) at each frequency, or a character string indicating the stated vocal effort corresponding to one of the standard standard speech spectrum levels ("normal", "raised", "loud", "shout"). Defaults to `speech="normal"` correspoding to the normal level of stated vocal effort.
`noise`	A numeric vector providing N_i', the equivalent noise spectrum level (in dB) at each frequency. If missing, defaults to -50 dB for each frequency.
`threshold`	A numeric vector providing T_i', the equivalent hearing threshold level (in dB) at each frequency. If missing, defaults to 0 dB for each frequency.
`loss`	A numeric vector providing J_i', the conductive hearing loss level (in dB) at each frequency. If missing, defaults to 0 dB for each frequency.
`freq`	Vector of frequencies for which `speech`, `noise`, `threshold`, and/or `loss` are specified. If `interpolate=TRUE`, `freq` must be specified. Otherwise, it must either match the required value for SII calculation method given by argument `method`, or be missing, in which case it will default to the values required for the specified method.

`method` A character string specifying the SII calculation method ("critical", "one-third octave", "equal-contributing", "octave")

`importance` Either a numeric vector providing F_i, the transfer function (importance weights) at each frequency, or a character string indicating which transfer function to employ ("SII", "NNS", "CID22", "NU6", "DRT", "ShortPassage", "SPIN", "CST"). Defaults to the standard SII transfer function, `importance="SII"`.

`interpolate` Logical flag indicating whether to interpolate from the provide measurement values and frequencies to those required by the specified method via linear interpolation on the log scale.

`prescription` A character string (e.g. `"NAL-R"`, `"Open-NL"`). If provided, the function will automatically calculate hearing aid insertion gain based on the specified prescriptive fitting rationale and add it to the speech and noise spectrum, calculating an Aided SII.

`desensitization`
Logical flag. If `TRUE`, applies the hearing loss desensitization factor (Ching et al., 2011; Johnson, 2013) to the calculation to account for suprathreshold distortion in impaired ears. Defaults to `FALSE` (Traditional ANSI S3.5 calculation).

`ldl` Numeric vector specifying Loudness Discomfort Levels (LDL) in dB HL for each frequency. If `NULL`, limits are estimated based on hearing thresholds.

`gender` Character string specifying patient gender ("male" or "female"). Used for prescriptive algorithms. Defaults to "male".

`experience` Character string specifying user hearing aid experience ("experienced", "new", "power"). Defaults to "experienced".

`config` Character string for fitting configuration ("bilateral" or "unilateral"). Defaults to "bilateral".

`age` Character string for patient age group (e.g., "adult", "child_0_5", "child_6_11", "child_12_23", "child_24_35", "child_36_59"). Used for RECD corrections and pediatric prescriptive modifications.

`age_years` Optional. Integer specifying exact age in years, overriding the categorical ‘age‘ bin if provided.

`age_months` Optional. Integer specifying exact age in months. Only applies to pediatric RECD scaling.

`coupling` Character string specifying the acoustic coupling / vent ("custom_occluded", "double_dome", "tulip_dome", "open_dome", "vent_1mm_solid", "vent_2mm_solid", "vent_3mm_solid", "vent_1mm_hollow", "vent_2mm_hollow", "vent_3mm_hollow"). Modifies low-frequency leakage and insertion gain targets.

`module` Character string for hearing aid module ("standard" or other specific module constraints).

`transducer` Character string specifying the audiometric transducer used for testing ("inserts" or "supra_aural"). Determines RETSPL and RECD corrections for SPLogram plots.

`custom_gain` Numeric vector specifying a custom insertion gain array (in dB) for each frequency. If provided along with `prescription="Custom"`, this gain is explicitly applied.

`object, x`	SII object
`digits`	Number of digits to display
`clinical`	Logical flag. If `TRUE`, plots a patient-friendly clinical SPLogram highlighting the audible speech area. If `FALSE` (the default), the behavior depends on whether the object is aided or unaided. For unaided objects, it plots a diagnostic interpolation graph. For aided objects (when a prescription was used), it automatically generates a 3-line Insertion Gain plot for 55, 65, and 75 dB SPL input levels (dynamically loading and scaling the true Normal and Loud LTASS spectra).
`legend`	Logical flag. If `TRUE` (the default), draws a legend on the plot.
`legend_only`	Logical flag. If `TRUE`, suppresses the main plot and only draws the legend in the center of the plotting window. Useful for multi-plot layouts.
`...`	Optional arguments to `print`, `summary`, and `plot` methods

Details

American National Standard ANSI S3.5-1997 ("Methods for Calculation of the Speech Intelligibility Index") defines a method for computing a physical measure that is highly correlated with the intelligibility of speech as evaluated by speech perception tests given a group of talkers and listeners. This measure is called the Speech Intelligibility Index, or SII. The SII is calculated from acoustical measurements of speech and noise.

The `sii` function implements ANSI S3.5-1997 as described in the standard, without any attempt to optimize the performance. The implementation does, however, include the extension for handling conductive hearing loss from Annex A (utilizing the optional `loss` argument), and for utilizing alternative band weights (i.e. transfer function) appropriate for differing message contents (e.g. types of speech) as described in Annex B or user-specified band weights (utilizing the optional argument `importance`).

Further, this implementation provides a mechanism for interpolating/extrapolating available measurements to those required for the specified calculation procedure. When `interpolate=TRUE`, required values for `speech`, `noise`, `threshold`, and `loss` will be computed using linear interpolation (of the log-scaled data). In this case, missing values may be provided and will be appropriately interpolated.

Prescriptive Fitting Rationales: If the `prescription` argument is provided, the function will dynamically calculate a frequency-specific hearing aid insertion gain and apply it to the speech and noise spectrum, calculating an Aided SII. The following rationales are supported:

- `"NAL-R"`: A classic linear fitting rationale designed by the National Acoustic Laboratories to maximize speech intelligibility for mild-to-moderate losses. Gain is a linear function of the pure-tone average (PTA) and frequency-specific thresholds.
- `"Open-NL"`: A completely transparent, open-source non-linear fitting algorithm developed specifically for this package. It was designed to serve as an open-source alternative to proprietary modern non-linear clinical targets (such as NAL-NL2 and DSL v5.0). The algorithm calculates insertion gain using the following explicit mathematical steps:
  1. **Conductive Component Separation**: If an Air-Bone Gap is present, the purely sensorineural component is isolated for WDRC compression. Linear gain representing 75% of the conductive loss is added at the end (consistent with NAL-NL2/Johnson).

2. **Dynamic Base Gain**: The base gain relies on a dynamic multiplier (ranging from 0.43 to 0.48) determined by user experience, combined with empirical NAL-R shaping constants. Severe losses receive a booster, while steep slopes receive a low-frequency penalty to avoid upward spread of masking.
3. **Dead Region Roll-offs**: High-frequency or low-frequency dead regions trigger a steep 30 dB/octave penalty beyond the viable boundaries to prevent acoustic distortion and feedback.
4. **Wide Dynamic Range Compression (WDRC)**: A dynamic compression ratio (ranging from 1.0 to 3.0) is applied symmetrically around a 65 dB SPL pivot point. The ratio scales dynamically based on the sensorineural threshold and measured Loudness Discomfort Levels (LDLs).
5. **Bandwidth Roll-off**: Empirical bandwidth roll-offs are applied to limit unnecessary low/high frequency amplification, with separate curves for adults vs. infants.
6. **Acoustic Coupling**: Adjustments are applied to the low frequencies depending on the specified venting, dome type, and acoustic seal.
7. **MPO Limits**: Output is capped by predictive NAL-SSPL90 limits, adjusting for conductive attenuation, with an absolute safety hard cap at 120 dB SPL at the cochlea and a hardware output limit of 135 dB SPL.

Value

The return value is an object of class SII, containing the following components:

`call` Function call used to generate the SII object

`orig` List containing original (pre-extrapolation) values for `freq`, `speech`, `noise`, `threshold`, and `loss`.

`speech`, `noise`, `threshold`, `loss`, and `freq`
Values used in calculations (extrapolated if necessary)

`unaided_speech` Original speech array before applying any prescriptive gain

`vocal_effort` String representing the stated vocal effort

`gain` Insertion gain array added to speech and noise when a prescription is used

`prescription` The fitting rationale used (if any)

`unaided_sii` Calculated SII value before applying the prescriptive gain (if a prescription was used)

`table` SII calculation worksheet, containing columns corresponding to both Table C.1 and C.2 in Annex C of the standard. Table columns are

- **Fi** Center frequency of SII band, Hz
- **E'i** Spectrum level of equivalent speech, dB
- **N'i** Spectrum level of equivalent noise, dB
- **T'i** Equivalent hearing threshold level, dB
- **Vi** Spectrum level for self-speech masking, dB
- **Bi** Larger of the specrum levels for equivalent noise and self-speech masking, dB
- **Ci** Slope per octave (doubling of frequency) of the upward spread of masking, dB/octave

- **Zi** Spectrum level for equivalent masking, dB
- **Xi** Spectrum level of internal noise, dB
- **X'i** Spectrum level of equivalent internal noise, dB
- **Di** Spectrum level for equivalent disturbance, dB
- **Ui** Spectrum level of standard speech for normal vocal effort, dB
- **Ji** Equivalent hearing threshold due to conductive hearing loss, dB
- **Li** Speech level distortion factor, dB
- **Ki** Temporary variable used in the calculation of the band auditability function
- **Ai** Band auditability function
- **Ii** Band importance function
- **IiAi** Product of the band importance function (Ii), and band auditability function(Ai)

`sii` Calculated SII value

Open-NL Prescription Rationale

DISCLAIMER: `Open-NL` is an untested, experimental fitting rationale. It is designed to mimic aspects of other generic non-linear fitting rationales (such as NAL-NL2 and DSL v5.0) strictly to promote open research, algorithmic transparency, and rapid iteration within the audiology community. It is not intended for clinical use.

When `prescription = "Open-NL"` is selected, the function calculates and applies this WDRC-optimized rationale to maximize the aided Speech Intelligibility Index.

The mathematical algorithm consists of:

1. **Minimal Hearing Loss (MHL) Bypass:** If `module == "mhl"` and $PTA_{.5,1,2,4k} \leq 25$ dB HL, WDRC is bypassed. Applies flat linear gain interpolated from $(f, G) = \{(250, 0), (500, 0), (1k, 3), (2k, 5), (4k, 5), (8k$ dB, with a 1.5 compression ratio for loud inputs.
2. **Base Anchor (65 dB SPL Input):** $g_{65} = \max(0, m \times HTL + C)$, where $m \in \{0.40, 0.45, 0.50\}$ depending on `experience`, and C interpolates arrays such as $\{-8, -1, +3, +1, 0, 0, 0, 0\}$ dB evaluated at standard audiometric frequencies.
3. **Audiometric Profile Corrections:**
   - *Steep Slope Knee:* For slopes > 30 dB/octave, penalizes the "knee" (500-1500 Hz) by up to 6 dB, and boosts frequencies ≥ 2000 Hz by up to 6 dB.
   - *Severe-Loss Booster (SLB):* Adds $\min(15, \max(0, HTL - 60) \times 0.5)$ dB. Tapered heavily in mid-frequencies and disabled in dead regions.
4. **High-Frequency Desensitization:** To prevent distortion, excess gain above a limit is compressed at a 2:1 ratio.
   - Adult $Limit = 30 + 0.4 \times \max(0, HTL - 60)$
   - Child $Limit = 40 + 0.5 \times \max(0, HTL - 60)$
5. **Roll-offs and Dead Regions:**
   - *Bandwidth Roll-off:* Multiplies g_{65}. Adults: interpolates $\{(250Hz, 0.7), (500, 1.0), ..., (6k, 0.8), (8k, 0.5)\}$. Children: 1.0 across all frequencies.
   - *Dead Regions:* Identifies HF Dead Regions ($HTL \geq 90$ at $f \geq 1k$) and LF Dead Regions ($HTL \geq 80$ at $f \leq 1k$). Applies a 30 dB/octave penalty beyond viable boundaries ($1.7 \times f_e$ for HF, $0.57 \times f_e$ for LF).

6. **Bi-directional WDRC:**
   - *Compression Ratio (CR):* $CR_{base} = 1 + \max(0, HTL - 20)/40$. For $HTL > 65$, CR reduces toward 1.0 in low frequencies. Overall bounded strictly to ≤ 1.5 for low frequencies (≤ 500 Hz) and up to 2.4 for high frequencies (≥ 3000 Hz).
   - *Compression Threshold (CT):* Interpolated from (HTL, CT). Adult: $\{(20HL, 30SPL), ..., (100, 45)\}$. Child: $\{(20, 25), ..., (100, 40)\}$.
   - *Aggressive MPO Defense:* For steeply sloping losses (> 15 dB difference) with low LDLs (< 100 dB SPL), the formula proactively defends against MPO collision. The CT is aggressively lowered by up to 10 dB, and the CR is forced up by an additional 0.05 per dB of LDL penalty.
   - *I/O Computation:* Calculates gain at CT (G_{CT}) scaling back from 65 dB SPL pivot. Applies linear gain below CT, and WDRC above CT.
7. **Empirical Adjustments and Smoothing:**
   - *Demographic Boosts:* Child Boost: Interpolates $(HTL, Boost) = \{(20, 5), (50, 5), (80, 2), (100, 1)\}$ dB. Gender: Female $= -1.5$ dB. Config: Unilateral $= +3.0$ dB. Experience: New users with $PTA > 40$ get up to -6.0 dB penalty.
   - *LDL Dynamic Range Mapping:* If `ldl` is provided, dynamic range is evaluated. For every 1 dB the measured LDL is lower than predicted, g_{65} is reduced by 0.2 dB and CR_{base} is increased by 0.02.
   - *Acoustic Coupling:* Subtracts vent leakage (e.g., Open Dome $= \{-35, -28, -15, -2, 0, 0\}$ dB).
   - Applies a 3-point moving average to the final gain array.
8. **SSPL90 MPO Limiting:**
   - $MPO_{heuristic} = 100 + 0.5 \times \max(0, HTL - 40)$.
   - $MPO_{safe} = LDL_{spl} - 5$.
   - $PTS_{safe\_limit} = 105 + 0.5 \times \max(0, HTL - 50)$ (110 for Children).
   - $MPO_{final} = \min(120, MPO_{heuristic}, MPO_{safe}, PTS_{safe\_limit})$.

Author(s)

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References

ANSI S3.5-1997, "American National Standard Methods for Calculation of the Speech Intelligibility Index" American National Standards Institute, New York.

Other software programs for calculating SII are available from https://sii.to/html/programs.html.

See Also

SII Constants: critical, and sic.critical

Examples

```
## Example C.1 from ANSI S3.5-1997 Annex C
sii.C1 <- sii(
              speech   = c(50.0, 40.0, 40.0, 30.0, 20.0,  0.0),
              noise    = c(70.0, 65.0, 45.0, 25.0,  1.0,-15.0),
              threshold= c( 0.0,  0.0,  0.0,  0.0,  0.0,  0.0),
              method="octave"
       )
sii.C1                         # rounded to 2 digits by default
print(sii.C1$sii, digits=20)  # full precision
summary(sii.C1)               # full details
plot(sii.C1)                  # plot
plot(sii.C1, clinical=TRUE)   # clinical SPLogram plot
## The value given in the Standard is $0.504$.



## Same calculation, but manually specify the frequencies
## and importance function, and use default for threshold

sii.C1 <- sii(
              speech   = c(50.0, 40.0, 40.0, 30.0, 20.0,  0.0),
              noise    = c(70.0, 65.0, 45.0, 25.0,  1.0,-15.0),
              method="octave",
              freq=c(250, 500, 1000, 2000, 4000, 8000),
       importance=c(0.0617, 0.1671, 0.2373, 0.2648, 0.2142, 0.0549)
       )
sii.C1

## Now perform the calculation using frequency weights for the Connected
## Speech Test (CST)
sii.CST <- sii(
               speech   = c(50.0, 40.0, 40.0, 30.0, 20.0,  0.0),
               noise    = c(70.0, 65.0, 45.0, 25.0,  1.0,-15.0),
               method="octave",
        importance="CST"
       )
round(sii.CST$table[,-c(5:7,13)],2)
sii.CST$sii

## Example C.2 from ANSI S3.5-1997 Annex C

sii.C2 <- sii(
              speech   = rep(54.0, 18),
              noise    = c(40.0, 30.0, 20.0, rep(0, 18-3) ),
              threshold= rep(0.0,  18),
              method="one-third"
              )
sii.C2$table[1:3,1:8]
sii.C2

## Interpolation example, for 8 frequencies using NU6 importance
```

```
## weight, default values for noise.
sii.left <- sii(
                speech="raised",
                threshold=c(25,25,30,35,45,45,55,60),
                freq=c(250, 500, 1000, 2000, 3000, 4000, 6000, 8000),
                method="critical",
                importance="NU6",
                interpolate=TRUE
                )
sii.left
```

Index